HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines on **AWS Lambda** and **S3** to process ~3,500 events per second, replacing a legacy batch job for feature engineering.
- Developed and deployed **Java Spring Boot** REST APIs for cross-service data exchange, incorporating request validation and circuit-breaker patterns for enhanced resilience.
- Engineered event-driven data processing workflows using **Python scripting** and **SQL (PostgreSQL)** to automate report generation for financial operations, reducing manual effort by 40%.
- Contributed to the design and implementation of scalable **AI data architecture patterns**, focusing on data accessibility and optimisation for machine learning experimentation.
- Prepared, cleansed, and transformed large datasets using **Python (pandas, NumPy)** and **SQL**, supporting AI use case development and feature engineering activities for enterprise stakeholders.
- Containerised data pipeline services using **Docker** and orchestrated deployments onto **Kubernetes (EKS)**, integrating with **Jenkins** for CI/CD automation and establishing resource limits for predictable performance.
- Implemented data governance, security, and monitoring controls within data pipelines, ensuring compliance with regulatory requirements and facilitating audit trail generation.

SKILLS

Cloud Platform – Azure Data Services, Microsoft Fabric, Databricks, Synapse, AWS (EC2, S3, Lambda, CloudWatch)

Programming Languages – Python, Java, SQL, R

Data Engineering & AI – Data Pipelines, ETL, Feature Engineering, Machine Learning Lifecycle, MLOps, Data Governance, AI Architecture, Generative AI

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Databases – PostgreSQL, MySQL, Oracle

Tools & Technologies – Docker, Kubernetes, Jenkins, Git, Jira, Splunk, Dynatrace

BI & Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling

PROJECTS

AI Data Pipeline Orchestration (HCL Technologies)

- Constructed robust **Python** data pipelines leveraging **AWS services (S3, Lambda)** to ingest and transform diverse datasets for AI model training, processing up to **3,000 events per second** during peak loads.
- Integrated **SQL (PostgreSQL)** queries within data pipelines for data validation and enrichment, ensuring data quality and consistency for downstream machine learning tasks.
- Containerised pipeline components with **Docker**, facilitating consistent deployment across development and production environments and improving deployment speed by **20%**.

Enterprise Feature Store Development (HCL Technologies)

- Designed and implemented REST APIs using **Java Spring Boot** to serve features for machine learning models, focusing on low-latency retrieval and data consistency.
- Developed data extraction and transformation scripts in **Python** to populate the feature store from various enterprise systems, supporting cross-system integrations.
- Collaborated with business analysts to define feature requirements, ensuring alignment between data engineering efforts and AI use case objectives.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)